

**[CS-466] Machine Learning
Spring 2026**

Announced date: 01/06/2026

Due Date: 29/06/2026

Total Marks = 04

Teacher Name: Ms. Tahmina Khan

Marks Obtained =

Complex Computing Problem

CLO #	CLO Statement	Blooms Taxonomy	GAs	Knowledge Profile Attributes
3	To be able to analyze different algorithms to improve the performance of machine learning models.	C4 (Analyze)	GA-3 (Problem Analysis)	D.4.1.1, D.4.1.2, D.4.1.3, D.4.1.7, D.4.1.9

Problem Statement of CCP:

To develop, implement, and evaluate deep learning models utilizing transfer learning to accurately detect and classify:

- **Tuberculosis (TB) from Chest X-Ray images** and
- **Brain Tumors from MRI scans**

Requirements:**1. Data Preprocessing:**

- Split each dataset into training (80%) and validation sets (20%).
- Normalize image pixel values between 0 and 1.
- Resize images to 256x256 pixels.

2. Transfer Learning:

- Use each of the following pre-trained models as a starting point:
 - ResNet50
 - DenseNet121
 - InceptionV3
 - VGG16
- Fine-tune the models by adjusting the following hyperparameters:
 - Learning rate (0.001, 0.01, 0.1)
 - Batch size (8, 16, 32)
 - Number of epochs (50)

3. Performance Evaluation:

- Calculate the following metrics for each model on the validation set:
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - AUC-ROC
 - -Confusion matrix

4. Comparative Analysis:

- Create a table comparing the performance metrics of each model for both datasets.
- Plot the ROC curves for each model.
- Discuss the strengths and weaknesses of each model and identify the best-performing model for each disease.

Deliverables:

- A Python script or Jupyter Notebook implementing the task.
- A table comparing the performance metrics of each model for both datasets.
- A plot (e.g., bar chart or ROC curve) visualizing the performance comparison.
- A brief report (4-5 pages) discussing the results, strengths, and weaknesses of each model, and concluding with the best-performing model for each disease.

Evaluation Criteria:

- Accuracy and efficiency of the transfer learning approach. –
- Effectiveness of hyperparameter tuning.
- Thoroughness of the comparative analysis.
- Clarity and organization of the deliverables.

Additional Requirements:

- Use techniques like data augmentation, regularization, or ensemble methods to improve model performance.
- Explore the use of transfer learning with different architectures
- Discuss the clinical implications and potential applications of the best-performing models in medical diagnosis.

COMPLEX COMPUTING PROBLEM:

Complex Computing Problem Details	Included: Yes Nature and details of Complex Computing Problem (CCP): It will be given in Assignment # 03. CCP will be based on CLO 3 "To be able to analyze different algorithms to improve the performance of machine learning models.". To solve the problem, students have to use in-depth knowledge related to the following concepts: - Transfer Learning - Normalization - Fine-tuning - Cross-validation - Performance Metrics - Hyperparameter tuning Attributes: D.4.1.1, D.4.1.2, D.4.1.3, D.4.1.7, D.4.1.9 D.4.1.1: Range of conflicting requirements D.4.1.2: Depth of analysis required D.4.1.3: Depth of knowledge required D.4.1.7: Consequences D.4.1.9: Requirement identification Assessment in: Assignment # 03
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